Controlled-Z (CZ) gate using the qutrit levels of a transmon pair

Quantum circuit electrodynamics and Quantum devices

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Presentation Outline

CLASSICAL SQUID

CLASSICAL Two Capacitively Coupled SQUIDs

QUANTUM Transmon

QUANTUM Two Capacitively Coupled SQUIDs

ROTATING FRAME

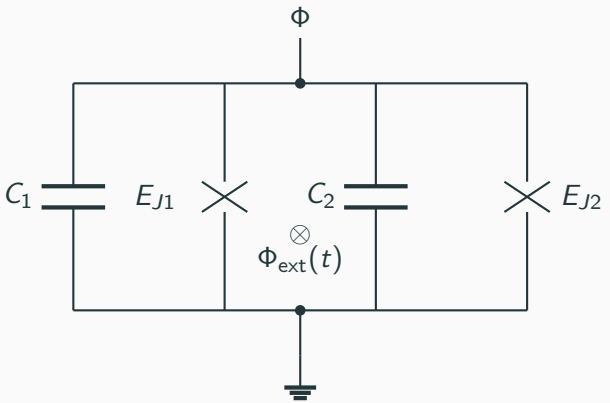
QUTRIT AND RWA

AVOIDED CROSSING

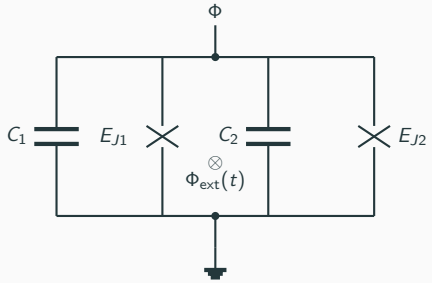
PULSE

CLASSICAL SQUID

SQUID Circuit



Definition of Flux Variables



$$\Phi_{C1} = -\Phi$$

$$\Phi_{J1} = \Phi$$

$$\Phi_{C2} = \Phi - \Phi_{\text{ext}}$$

$$\Phi_{J2} = \Phi - \Phi_{\text{ext}}$$

Lagrangian

Kinetic Energy (Capacitive)

$$T = \frac{1}{2} C_1 \dot{\Phi}^2 + \frac{1}{2} C_2 (\dot{\Phi} - \dot{\Phi}_{\text{ext}})^2 \tag{1}$$

Potential Energy (Josephson)

$$U = E_{J1} \left[1 - \cos\left(\frac{2\pi\Phi}{\Phi_0} \right) \right] + E_{J2} \left[1 - \cos\left(\frac{2\pi(\Phi - \Phi_{\text{ext}})}{\Phi_0} \right) \right] \tag{2}$$

Transition to the Symmetric Gauge

We introduce a transformation of the flux coordinates to transition to the symmetric gauge. Applying the substitution:

$$\Phi \to \Phi + \frac{\Phi_{\text{ext}}}{2} \quad \Longrightarrow \quad \dot{\Phi} \to \dot{\Phi} + \frac{\dot{\Phi}_{\text{ext}}}{2}$$

The expressions for the kinetic and potential energy become:

Kinetic Energy

$$T = \frac{1}{2} C_1 \left(\dot{\Phi} + \frac{\dot{\Phi}_{\text{ext}}}{2} \right)^2 + \frac{1}{2} C_2 \left(\dot{\Phi} - \frac{\dot{\Phi}_{\text{ext}}}{2} \right)^2 \tag{3}$$

Potential Energy

$$U = E_{J1} \left[1 - \cos \left(\frac{2\pi}{\Phi_0} \left(\Phi + \frac{\Phi_{\text{ext}}}{2} \right) \right) \right] + E_{J2} \left[1 - \cos \left(\frac{2\pi}{\Phi_0} \left(\Phi - \frac{\Phi_{\text{ext}}}{2} \right) \right) \right] \tag{4}$$

Simplification: Symmetric Case

We consider a symmetric device, defining the total capacitance and Josephson energy:

$$C = C_1 = C_2 \implies C_{\mathrm{TOT}} = C_1 + C_2 = 2C$$
$$E_J = E_{J1} = E_{J2} \implies E_J^{\mathrm{TOT}} = E_{J1} + E_{J2} = 2E_J$$

Kinetic and Potential Energies

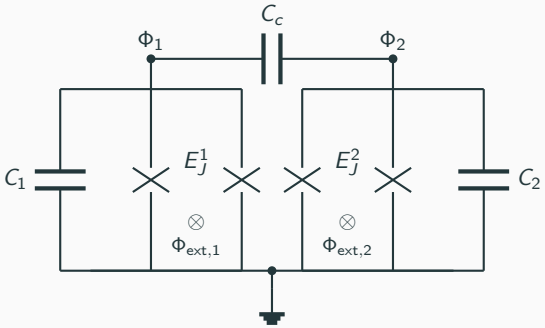
$$T = \frac{1}{2} C_{\mathrm{TOT}} \dot{\Phi}^2 + \frac{C_{\mathrm{TOT}}}{8} \dot{\Phi}_{\mathrm{ext}}^2$$
$$U = E_J^{\mathrm{TOT}} \left[1 - \cos\left(\frac{2\pi}{\Phi_0} \Phi \right) \cos\left(\pi \frac{\Phi_{\mathrm{ext}}(t)}{\Phi_0} \right) \right]$$

The Lagrangian of the system, defined as $L = T - U$, is therefore:

$$L = \frac{1}{2} C_{\mathrm{TOT}} \dot{\Phi}^2 - E_J^{\mathrm{TOT}} \left[1 - \cos\left(\frac{2\pi}{\Phi_0} \Phi \right) \cos\left(\pi \frac{\Phi_{\mathrm{ext}}(t)}{\Phi_0} \right) \right] \tag{5}$$

CLASSICAL Two Capacitively Coupled SQUIDs

Two Capacitively Coupled SQUIDs



Lagrangian of the Coupled System

$$L = \sum_{k=1}^{2} \left[\frac{1}{2} C_k \dot{\Phi}_k^2 - E_J^k \left(1 - \cos\left(\pi \frac{\Phi_{\text{ext},k}(t)}{\Phi_0} \right) \cos\left(\frac{2\pi}{\Phi_0} \Phi_k \right) \right) \right] + \frac{1}{2} C_c (\dot{\Phi}_1 - \dot{\Phi}_2)^2$$

Lagrangian with Capacitance Matrix

We can rewrite the kinetic term of the Lagrangian in a compact matrix form. We introduce the nodal flux vector and the capacitance matrix of the system:

$$\mathbf{\Phi} = \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix} \qquad \mathrm{C} = \begin{pmatrix} C_1 + C_c & -C_c \\ -C_c & C_2 + C_c \end{pmatrix}$$

Lagrangian in Matrix Form

$$L = \frac{1}{2}\dot{\mathbf{\Phi}}^T \mathrm{C} \dot{\mathbf{\Phi}} - \sum_{k=1}^{2} E_J^k \left[1 - \cos\left(\pi \frac{\Phi_{\mathrm{ext},k}(t)}{\Phi_0} \right) \cos\left(\frac{2\pi}{\Phi_0} \Phi_k \right) \right] \tag{6}$$

Transition to the Hamiltonian

We can transition to the Hamiltonian formalism by defining the charges (conjugate momenta) associated with the nodal fluxes:

$$\mathsf{Q} = \frac{\partial L}{\partial \dot{\mathbf{\Phi}}} = \mathsf{C}\dot{\mathbf{\Phi}}$$

System Hamiltonian

$$\mathcal{H} = -\frac{1}{2}\mathsf{Q}^T\mathsf{C}^{-1}\mathsf{Q} + U \tag{7}$$

Hamiltonian Expansion

The inverse of the capacitance matrix is explicitly calculated as:

$$C^{-1} = \frac{1}{\Sigma} \begin{pmatrix} C_2 + C_c & C_c \\ C_c & C_1 + C_c \end{pmatrix}$$

where the denominator factor is given by:

$$\Sigma = C_1 C_c + C_2 C_c + C_1 C_2$$

Explicit System Hamiltonian

$$\mathcal{H} = \frac{1}{2} \frac{C_2 + C_c}{\Sigma} Q_1^2 + \frac{1}{2} \frac{C_1 + C_c}{\Sigma} Q_2^2 + \frac{C_c}{\Sigma} Q_1 Q_2 + U \tag{8}$$

Weak Coupling Approximation

In the **Weak Coupling** limit, i.e., when the coupling capacitance is negligible compared to the capacitances of the individual SQUIDs:

$$C_c \ll C_1 \quad \text{and} \quad C_c \ll C_2$$

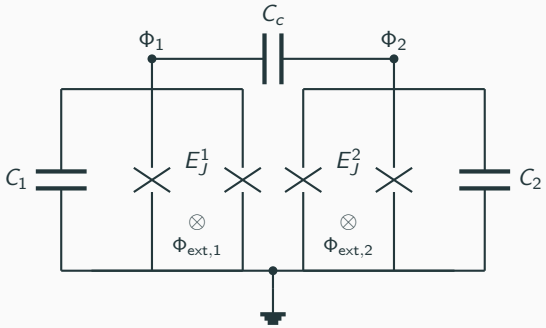
The normalization term Σ simplifies as follows:

$$\Sigma = C_1 C_c + C_2 C_c + C_1 C_2 = C_1 C_2 \left(1 + \frac{C_c}{C_2} + \frac{C_c}{C_1}\right) \approx C_1 C_2$$

Under these assumptions, the coefficients of the Hamiltonian become:

- $\frac{C_2 + C_c}{\Sigma} \approx \frac{C_2}{C_1 C_2} = \frac{1}{C_1}$
- $\frac{C_1 + C_c}{\Sigma} \approx \frac{C_1}{C_1 C_2} = \frac{1}{C_2}$
- Interaction term: $\frac{C_c}{\Sigma} \approx \frac{C_c}{C_1 C_2}$

Two Capacitively Coupled SQUIDs



Hamiltonian with Weak Coupling

$$H = \left[\sum_{k=1}^{2} \frac{Q_k^2}{2C_k} + E_J^k \left(1 - \cos\left(\pi \frac{\Phi_{\text{ext},k}(t)}{\Phi_0}\right) \cos\left(\frac{2\pi}{\Phi_0}\Phi_k\right)\right)\right] + Q_1 Q_2 \frac{C_c}{C_1 C_2}$$

QUANTUM Transmon

Quantization of the Transmon Hamiltonian

We now study the Transmon equation by quantizing its classical Hamiltonian:

$$H = \frac{Q^2}{2C} + E_J \left[1 - \cos\left(\frac{2\pi}{\Phi_0}\Phi\right)\right]$$

Quantization Rules

$$Q \quad \longrightarrow \quad \hat{q} = -i\hbar \frac{2\pi}{\Phi_0} \frac{\partial}{\partial \varphi}$$

$$\frac{2\pi}{\Phi_0}\Phi \quad \longrightarrow \quad \hat{\varphi}$$

By applying these rules, we obtain the quantized Transmon Hamiltonian:

$$\hat{H} = \frac{\hat{q}^2}{2C} + E_J \left[1 - \cos(\hat{\varphi})\right] \tag{10}$$

Transmon Approximation: Perturbative Regime

The Schrödinger equation associated with this Hamiltonian translates into Mathieu equations. However, for Transmon-type qubits, we operate in a specific regime where the Josephson energy dominates the charging energy: $\frac{E_J}{E_C} \gg 50$

In this limit, phase fluctuations are small enough to allow a **perturbative approach**. Expanding the $\cos(\hat{\varphi})$ term in a Taylor series, we obtain:

Hamiltonian Expansion

$$\hat{H} \approx \frac{\hat{q}^2}{2C} + E_J \left(1 - 1 + \frac{\hat{\varphi}^2}{2} - \frac{\hat{\varphi}^4}{24}\right)$$

By grouping the terms, we separate the harmonic oscillator part from the anharmonic perturbation:

$$\hat{H} \approx \underbrace{\frac{\hat{q}^2}{2C} + \frac{E_J}{2}\hat{\varphi}^2}_{\hat{H}_0} \underbrace{- \frac{E_J}{24}\hat{\varphi}^4}_{\hat{V}} \tag{11}$$

Second Quantization: Ladder Operators

This form allows us to solve the system using creation and annihilation operators $(\hat{a}^\dagger, \hat{a})$, treating $\hat{V}$ as a perturbation on $\hat{H}_0$.

We define the flux and charge operators as a function of the creation and annihilation operators, introducing the zero-point fluctuations (ZPF):

Operators

$$\hat{\Phi} = \Phi_{\text{ZPF}}(\hat{a}^\dagger + \hat{a})$$

$$\hat{q} = iq_{\text{ZPF}}(\hat{a}^\dagger - \hat{a})$$

ZPF Amplitudes

$$\Phi_{\text{ZPF}} = \sqrt{\frac{\hbar Z_0}{2}}$$

$$q_{\text{ZPF}} = \sqrt{\frac{\hbar}{2Z_0}}$$

Transmon Hamiltonian

By substituting these definitions into the Hamiltonian expansion, we obtain:

Reformulated Hamiltonian

$$\hat{H}_0 = \hbar\omega_p \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) \qquad \text{with} \quad \hbar\omega_p = \sqrt{8E_C E_J}$$

$$\hat{V} = -\frac{E_C}{12}(\hat{a}^\dagger + \hat{a})^4 = -\frac{E_C}{2}(\hat{a}^\dagger \hat{a}^\dagger \hat{a}\hat{a} + 2\hat{a}^\dagger \hat{a} + \frac{1}{2})$$

QUANTUM Two Capacitively Coupled SQUIDs

Hamiltonian of Two Coupled Transmons

The total Hamiltonian of the system composed of two capacitively coupled Transmon qubits can be decomposed as:

$$\hat{H} = \hat{H}_1 + \hat{H}_2 + \hat{H}_{\text{int}} \tag{12}$$

$$\hat{H}_k(t) = \hbar\omega_p(t)\left(\hat{a}_k^\dagger \hat{a}_k + \frac{1}{2}\right) - \frac{E_{Ck}}{2}\left(\hat{a}_k^\dagger \hat{a}_k^\dagger \hat{a}_k \hat{a}_k + 2\hat{a}_k^\dagger \hat{a}_k + \frac{1}{2}\right)$$

$$\hat{H}_{\text{int}} = -g\hbar(\hat{a}_1^\dagger - \hat{a}_1)(\hat{a}_2^\dagger - \hat{a}_2)$$

where the coupling strength is:

$$g = \frac{\sqrt{2}}{\hbar}\frac{C_c}{\sqrt{C_1 C_2}}(E_{J1}E_{C1}E_{J2}E_{C2})^{1/4} \tag{13}$$

External Flux Dependence and Adiabatic Approximation

We introduce the dependence of the plasma frequency on the external magnetic flux $\Phi_{\text{ext}}(t)$ only a posteriori:

Modulated Plasma Frequency

$$\omega_p(t) = \sqrt{8E_C E_J \cos\left(\frac{\pi\Phi_{\text{ext}}(t)}{\Phi_0}\right)} \tag{14}$$

This procedure is formally correct if the time taken to vary the external flux is extremely slow compared to the time scale of the quantum dynamics (**Adiabatic Approximation**):

- **Condition:** $\tau_{\text{ext}} \gg \tau_p = \tau_p = \frac{2\pi}{\omega_p} \sim$ order of nanoseconds

In this regime, the system "sees" the flux as a quasi-static constant at every instant of its coherent evolution.

Elimination of the Energy Shift

We reorganize the Hamiltonian by neglecting additive constants (such as zero-point energy) and grouping the linear terms into $\hat{n}_k = \hat{a}_k^\dagger \hat{a}_k$.

The total Hamiltonian $\hat{H} = \hat{H}_1 + \hat{H}_2(t) + \hat{H}_{\text{int}}$ is explicitly rewritten as:

Hamiltonian Components

$$\hat{H}_1 = (\hbar\omega_p^1 - E_C^1)\hat{a}_1^\dagger \hat{a}_1 - \frac{E_C^1}{2}\hat{a}_1^\dagger \hat{a}_1^\dagger \hat{a}_1 \hat{a}_1$$

$$\hat{H}_2(t) = (\hbar\omega_p^2(t) - E_C^2)\hat{a}_2^\dagger \hat{a}_2 - \frac{E_C^2}{2}\hat{a}_2^\dagger \hat{a}_2^\dagger \hat{a}_2 \hat{a}_2$$

$$\hat{H}_{\text{int}} = -\hbar g \left(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_1 \hat{a}_2^\dagger - \hat{a}_1^\dagger \hat{a}_2^\dagger - \hat{a}_1 \hat{a}_2\right)$$

ROTATING FRAME

Transition to the Rotating Frame

First, we define the effective transition frequencies of the two qubits:

$$\omega_1 = \omega_p^1 - \frac{E_C^1}{\hbar} \qquad \omega_2(t) = \omega_p^2(t) - \frac{E_C^2}{\hbar}$$

Unitary Transformation

We switch to a rotating frame of reference via the unitary operator $\hat{U}$. The transformed Hamiltonian is given by:

$$\hat{H}_{\text{rot}} = \hat{U}\hat{H}\hat{U}^\dagger - i\hbar\hat{U}\dot{\hat{U}}^\dagger \tag{15}$$

We choose to rotate the entire system at the frequency of the first qubit:

$$\hat{U} = \exp\left[i\omega_1 t\left(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2\right)\right]$$

Transition to the Rotating Frame

We explicitly calculate the derivative to evaluate the compensation term:

$$\dot{\hat{U}}^\dagger = -i\omega_1 \left(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2 \right) \hat{U}^\dagger$$

$$\tilde{H} \equiv -i\hbar \hat{U} \dot{\hat{U}}^\dagger = -\hbar\omega_1 \left(\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2 \right)$$

We recall the transformation properties of the creation and annihilation operators under the action of the unitary operator $\hat{U}$:

$$\hat{U} \hat{a}_k \hat{U}^\dagger = \hat{a}_k e^{-i\omega_1 t}$$

$$\hat{U} \hat{a}_k^\dagger \hat{U}^\dagger = \hat{a}_k^\dagger e^{i\omega_1 t}$$

Transition to the Rotating Frame

By applying these relations to the interaction term $\hat{H}_{\text{int}}$, the cross products $\hat{a}_1^\dagger \hat{a}_2$ and $\hat{a}_1 \hat{a}_2^\dagger$ lose their time dependence, while the simultaneous creation and annihilation terms acquire rapidly oscillating phases:

Interaction in the Rotating Frame

$$\hat{H}_{\text{int}}^{\text{rot}} = -\hbar g \left(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_1 \hat{a}_2^\dagger - \hat{a}_1^\dagger \hat{a}_2^\dagger e^{2i\omega_1 t} - \hat{a}_1 \hat{a}_2 e^{-2i\omega_1 t} \right) \tag{16}$$

Rotating Frame Hamiltonian: Detuning

By substituting the terms and applying the rotation, we obtain the final expression of the Hamiltonian in the rotating frame at frequency ω_1:

Rotating Frame Hamiltonian

$$\hat{H}_{\text{rot}} = -\frac{E_{C1}}{2}\hat{a}_1^\dagger\hat{a}_1^\dagger\hat{a}_1\hat{a}_1 + \hbar\Delta(t)\hat{a}_2^\dagger\hat{a}_2 - \frac{E_{C2}}{2}\hat{a}_2^\dagger\hat{a}_2^\dagger\hat{a}_2\hat{a}_2 + \hat{H}'_{\text{int}} \tag{17}$$

We introduced the **detuning** $\Delta(t)$, which represents the transition frequency difference between the two qubits:

$$\hbar\Delta(t) = \hbar(\omega_2(t) - \omega_1) = \hbar\omega_p^2(t) - E_{C2} - \hbar\omega_p^1 + E_{C1} \tag{18}$$

QUTRIT AND RWA

Projection into the Truncated Space (3 Levels)

If we project the system into a truncated space up to the third energy level ($|0\rangle, |1\rangle, |2\rangle$), the creation and annihilation operators in the three-level basis become:

$$\hat{a} = |0\rangle\langle 1| + \sqrt{2}|1\rangle\langle 2| \qquad \hat{a}^\dagger = |1\rangle\langle 0| + \sqrt{2}|2\rangle\langle 1|$$

System Hamiltonian (3 Levels)

$$\hat{H}_{\text{rot}} = -E_{C1}|2\rangle_1\langle 2|_1 + \hbar\Delta(t)|1\rangle_2\langle 1|_2 + (\hbar\Delta(t) - E_{C2})\,|2\rangle_2\langle 2|_2 + \hat{H}'_{\text{int}} \qquad (19)$$

$$\hat{H}'_{\text{int}} = \hbar g\Big[|1,0\rangle\langle 0,1| + \sqrt{2}|2,0\rangle\langle 1,1| + \sqrt{2}|1,1\rangle\langle 0,2| + 2|2,1\rangle\langle 1,2| + \text{h.c.}\Big]$$

$$-\hbar g\Big[[|1,1\rangle\langle 0,0| + \sqrt{2}|2,1\rangle\langle 1,0| + \sqrt{2}|1,2\rangle\langle 0,1| + 2|2,2\rangle\langle 1,1|]e^{2i\omega_1 t} + \text{h.c.}\Big]$$

Two-Qutrit Matrix Representation

By defining the rapidly oscillating phase term $\Omega = e^{2i\omega_1 t}$, we can construct the full matrix of the Hamiltonian $\hat{H}_{\text{rot}}$ in the tensor basis $\{|00\rangle, |01\rangle, \ldots, |22\rangle\}$:

	$\|00\rangle$	$\|01\rangle$	$\|02\rangle$	$\|10\rangle$	$\|11\rangle$	$\|12\rangle$	$\|20\rangle$	$\|21\rangle$	$\|22\rangle$
$\langle 00\|$	0	0	0	0	$-g\Omega^*$	0	0	0	0
$\langle 01\|$	0	$\Delta(t)$	0	g	0	$-\sqrt{2}g\Omega^*$	0	0	0
$\langle 02\|$	0	0	$2\Delta(t) - E_{C2}$	0	$\sqrt{2}g$	0	0	0	0
$\langle 10\|$	0	g	0	0	0	0	0	$-\sqrt{2}g\Omega^*$	0
$\langle 11\|$	$-g\Omega$	0	$\sqrt{2}g$	0	$\Delta(t)$	0	$\sqrt{2}g$	0	$-2g\Omega^*$
$\langle 12\|$	0	$-\sqrt{2}g\Omega$	0	0	0	$2\Delta(t) - E_{C2}$	0	$2g$	0
$\langle 20\|$	0	0	0	0	$\sqrt{2}g$	0	$-E_{C1}$	0	0
$\langle 21\|$	0	0	0	$-\sqrt{2}g\Omega$	0	$2g$	0	$\Delta(t) - E_{C1}$	0
$\langle 22\|$	0	0	0	0	$-2g\Omega$	0	0	0	$2\Delta(t) - E_{C1} - E_{C2}$

Rotating Wave Approximation (RWA)

If the coupling strength is sufficiently weak compared to the rotation frequency ($g \ll 2\omega_1$), it is possible to apply the **Rotating Wave Approximation (RWA)**.

In this regime, all counter-rotating terms (proportional to Ω and Ω^*) oscillate too rapidly to affect the dynamics and can be eliminated:

Interaction in the Rotating Frame with RWA

$$\hat{H}_{\text{rot}}^{\text{RWA}} = -\frac{E_{C1}}{2}\hat{a}_1^\dagger\hat{a}_1^\dagger\hat{a}_1\hat{a}_1 + \hbar\Delta(t)\hat{a}_2^\dagger\hat{a}_2 - \frac{E_{C2}}{2}\hat{a}_2^\dagger\hat{a}_2^\dagger\hat{a}_2\hat{a}_2 - \hbar g\left(\hat{a}_1^\dagger\hat{a}_2 + \hat{a}_1\hat{a}_2^\dagger\right) \tag{20}$$

RWA matrix

The two-qutrit matrix becomes:

	$\|00\rangle$	$\|01\rangle$	$\|02\rangle$	$\|10\rangle$	$\|11\rangle$	$\|12\rangle$	$\|20\rangle$	$\|21\rangle$	$\|22\rangle$
$\langle 00\|$	0	0	0	0	0	0	0	0	0
$\langle 01\|$	0	$\Delta(t)$	0	g	0	0	0	0	0
$\langle 02\|$	0	0	$2\Delta(t) - E_{C2}$	0	$\sqrt{2}g$	0	0	0	0
$\langle 10\|$	0	g	0	0	0	0	0	0	0
$\langle 11\|$	0	0	$\sqrt{2}g$	0	$\Delta(t)$	0	$\sqrt{2}g$	0	0
$\langle 12\|$	0	0	0	0	0	$2\Delta(t) - E_{C2}$	0	$2g$	0
$\langle 20\|$	0	0	0	0	$\sqrt{2}g$	0	$-E_{C1}$	0	0
$\langle 21\|$	0	0	0	0	0	$2g$	0	$\Delta(t) - E_{C1}$	0
$\langle 22\|$	0	0	0	0	0	0	0	0	$2\Delta(t) - E_{C1} - E_{C2}$

AVOIDED CROSSING

CZ Gate Realization: Avoided Crossing

At this point, to realize the **Controlled-Z (CZ)** gate, we need to apply a phase of -1 to the $|11\rangle$ state.

This is possible thanks to the phenomenon of **avoided crossing** with the auxiliary level $|02\rangle$.

Hamiltonian in the Subspace
The reduced Hamiltonian describing the isolated dynamics in this $\{|11\rangle, |02\rangle\}$ subspace is:

$$H_{sub} = \begin{pmatrix} \Delta & \sqrt{2}g \\ \sqrt{2}g & 2\Delta - E_{C2} \end{pmatrix} \tag{21}$$

Unitary Evolution at the Degeneracy Point

If the diagonal elements of the subspace Hamiltonian are degenerate ($\Delta = 2\Delta - E_{C2} \Rightarrow \Delta = E_{C2}$), the unitary evolution matrix can be written as:

Evolution Matrix in the Subspace

$$U_{\text{sub}}(t) = \exp\left(-i\frac{\Delta + (2\Delta - E_{C2})}{2}t\right) e^{-i\sqrt{2}g\sigma_x t} \tag{22}$$

The first exponential term produces a **global phase** $e^{i\varphi}$ common to the entire $\{|11\rangle, |02\rangle\}$ subspace.

Rotation

The second term, proportional to the Pauli matrix σ_x, describes the coherent rotation governing the population exchange between the states.

$$e^{-i\sqrt{2}g\sigma_x t} = \begin{pmatrix} \cos(\sqrt{2}gt) & -i\sin(\sqrt{2}gt) \\ -i\sin(\sqrt{2}gt) & \cos(\sqrt{2}gt) \end{pmatrix} \tag{23}$$

CZ Gate Implementation: Square Pulse

Let's assume we apply an external magnetic flux with a **square pulse** profile:

$$\Phi_{\text{ext}}(t) = \begin{cases} 0 & t < 0 \\ \bar{\Phi}_{\text{ext}} & 0 \leq t \leq T \\ 0 & t > T \end{cases} \tag{24}$$

The control problem reduces to finding the optimal pair of parameters $(\bar{\Phi}_{\text{ext}}, T)$:

- **Choice of $\bar{\Phi}_{\text{ext}}$:** It is determined by imposing the **degeneracy** condition in the subspace ($\Delta = E_{C2}$), placing the system at the maximum coupling point.
- **Choice of T:** It is determined by requiring that the phase accumulated by the $|11\rangle$ state relative to the other computational states is exactly equal to π (a factor of -1).

Optimal Parameters for the CZ Gate

By applying the conditions discussed in the square pulse protocol, we can analytically derive the exact values of the amplitude and duration.

$$\frac{\bar{\Phi}_{\text{ext}}}{\Phi_0} = \frac{1}{\pi} \arccos\left(\left[\frac{\omega_p^1 + 2E_{C2} - E_{C1}}{\omega_p^2}\right]^2\right) \tag{25}$$

$$T = \frac{\pi}{g\sqrt{2}} \tag{26}$$

PULSE

Adiabaticity Limits and Smoothed Pulses

To respect the adiabaticity limits and prevent unwanted diabatic transitions, it is not physically possible to use an ideal *square-pulse* flux. We therefore adopt a function with a "smooth" ramp that starts and ends at zero.

A commonly used profile is based on the error function:

$$\Phi_{\text{ext}}(t) = \frac{\Psi_{\text{ext}}}{2}\left[\text{erf}\left(\frac{t-3\sigma}{\sigma}\right) - \text{erf}\left(\frac{t-T+3\sigma}{\sigma}\right)\right] \tag{27}$$

The ramp time variable σ is added to the parameter space.

The amplitude Ψ_{ext} remains equal to the analytically calculated one (ensuring perfect degenerate coupling). The duration T, however, must be longer: since the external flux does not instantly reach the target value, the phase accumulation speed is reduced during the pulse transients.

Spurious Phases and Gate Correction

During the flux modulation to achieve the -1 phase on the $|11\rangle$ level (via *avoided crossing*), the system inevitably accumulates **spurious dynamical phases**: $\varphi = \frac{1}{\hbar}\int_0^T \Delta(t)dt$.

In the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the Hamiltonian operator is:

$$H(t) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \Delta(t) & g & 0 \\ 0 & g & 0 & 0 \\ 0 & 0 & 0 & \Delta(t) \end{pmatrix}$$

Spurious Phases and Gate Correction

The generic evolution operator $U(T)$ will therefore be:

$$U(T) = \begin{pmatrix} e^{i\varphi_0} & 0 & 0 & 0 \\ 0 & e^{i\varphi_1} & 0 & 0 \\ 0 & 0 & e^{i\varphi_0} & 0 \\ 0 & 0 & 0 & e^{i(\varphi_1+\pi)} \end{pmatrix} = \mathrm{CZ} \cdot \begin{pmatrix} e^{i\varphi_0} & 0 & 0 & 0 \\ 0 & e^{i\varphi_1} & 0 & 0 \\ 0 & 0 & e^{i\varphi_0} & 0 \\ 0 & 0 & 0 & e^{i\varphi_1} \end{pmatrix}$$

$$= e^{i\varphi_0}\mathrm{CZ}\,(\mathbb{I} \otimes P(\varphi_1 - \varphi_0))$$

So one can apply single-qubit pulses to exactly cancel the single-qubit phases.

$$\mathrm{CZ} = U(T)e^{-i\varphi_0}\,(\mathbb{I} \otimes P(\varphi_0 - \varphi_1)) \tag{28}$$